

Aaliyah Pirani

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EDUCATION

University of Toronto, St. George Campus

Graduating April 2028

Honours Bachelor of Science in Computer Science

GPA: 4.0/4.0

- Relevant Coursework: Software Design, DSA, Databases, Probability/Statistics, Computer Organization
- Awards: TD Community Leadership (\$70,000), CS Research Grant (\$12,000), Academic Awards (\$16,500)

PROFESSIONAL EXPERIENCE

Research Lead

May 2026 – Present

Medical Computer Vision and Robotics Lab, University of Toronto

Toronto, ON

- Using **NVIDIA Warp** and **Newton Physics Engine's** native solver APIs, ported an external finite element method (FEM) deformable solver so soft bodies can be simulated in the same engine as Newton's built-in solvers
- Integrated self-collision and rigid-soft collision in **Python** through **coupling callbacks** that apply contact forces from external FEM solve back to native rigid bodies, allowing for interaction of rigid and soft bodies in simulation
- Evaluating FEM alongside existing vertex block descent solver, to assess viability for **RL robot policy-training**

Software Developer

May 2026 – Sep. 2026

Centre for Diverse Learners

Burlington, ON

- Designed and shipped an internal **intake app**, replacing email threads and separate registration forms with a single automated workflow for tracking **300+ annual registrations** (Microsoft PowerApps, PowerAutomate)

Data Risk Intern

May 2025 – Aug. 2025

Toronto-Dominion Bank

Remote

- Built a Microsoft PowerApp for **cross-departmental intake and handoff**, modeled data in Power Query to model the data and PowerBI dashboards for completion metrics, significantly reducing workload internally

PROJECTS

BearWithMe | *Python, Node.js, Raspberry Pi*

Nov. 2025

- Embedded a voice tutor on a Raspberry Pi with **mic capture and speaker** that called Azure Speech API for pronunciation scoring, **Gemini API** for feedback and progress and **ElevenLabs TTS** to synthesize spoken replies
- Deployed a **Node.js** web app that pulled Gemini-generated reports so caregivers can remotely review progress

SnackOverflow | *Java, MongoDB, Git*

Nov. 2025 – Dec. 2025

- Collaborated with five others on a **Java Swing** desktop recipe tracker, using **clean architecture** and GitHub pull requests for smooth collaboration when working independently on the app
- Stored user data (recipes, bookmarks, etc.) in **MongoDB** and pulled new recipes using **Spoonacular API**

LEADERSHIP EXPERIENCE

Teaching Assistant

Sep. 2026 – Present

Department of Statistical Sciences, University of Toronto

Toronto, ON

- Ran tutorials and office hours for **introductory statistics and data science**, creating a welcoming environment
- Graded assignments/tests and invigilated exams, giving students consistent feedback in alignment with markscheme

Vice President

Sep. 2025 – Present

University of Toronto Robotics Association

Toronto, ON

- Acted as **primary liaison** between the administrative team and 6 robotics subteams, facilitating biweekly progress meetings, troubleshooting **technical and organizational** issues and ensuring alignment across subteams
- Formerly manager of Sumo, leading **CAD design**, **Arduino** wiring and **coding workshops** for autonomous bot

TECHNICAL SKILLS

Languages & Libraries: Python, Java, C, R, JSON, pandas, NumPy, PyTorch, NVIDIA Warp, scikit-learn

Software/Tools: Git, MATLAB, LaTeX, MongoDB, Node.js, MuJoCo, Newton, Arduino, SolidWorks, Microsoft PowerBI, PowerAutomate, PowerApps, Raspberry Pi